

EVERY CAR GETS ITS OWN GARAGE. EVERY CARD STAYS ON SHOW.

PITLANE is a 3D-printed wall system for carded die-cast cars. One 40 cm strip holds up to 7 cars, and each card sits in its own swing-out ledge. Lift a ledge, swing it open, slide the card in, swing it shut. The cards step out 7 mm per row, so every name stays visible and no card covers another.

The pitch (30 seconds): "Collectors keep their best cars carded, then hide them in boxes or crush them on peg hooks. PITLANE gives every card its own swing-out garage on the wall. You can see every name, take any one card out without touching the others, and add racks as your collection grows. It snaps together with no tools. One 40 cm strip holds up to 7 cars, for about the price of one mainline car per slot."



Real printed prototype loaded with real carded cars (designer's photo).



Current design: 7 cars on one 40 cm strip (CAD render, cards shown as blanks).

WHAT'S IN THE BOX (7-CAR KIT)

1x wall strip, 40 cm

One piece, or two halves that lock with a C-interlock. Fixed with 4 screws. Hook-lip pairs every 54 mm.

7x hinge clips

Each clip double-hooks behind two lips, and two solid bumps click into notches in the strip (no springs). Each one sits 7 mm deeper than the one below. No tools.

7x swing ledges

8 mm pin running the full height, a 0° stop, a click detent and 14 mm corner posts that keep the card upright.

WHY IT WINS

- **Grab one, leave the rest:** other displays make you unhook the cards in front. Here each card has its own door.
- **Names stay readable:** a 1 mm front lip and stepped rows mean nothing covers the card title.
- **Built for real prints:** every part prints without supports, and every lock was redesigned after real test prints failed.
- **Grows with the collection:** buy the strip once and add racks for ₹199 each.

MARKET

Adult collectors now drive the die-cast market worldwide (an estimated 58% of revenue, a \$4.2–5.8 billion market growing 5–6% a year). India's collector scene is young and growing fast: mainline cars sell for roughly ₹200–450 each online, and collectors who keep cars **carded** (mint in package) have few wall options made for cards. That gap is what PITLANE fills.

COMPETITION AND PRICES IN INDIA (OCT 2026)

Product	What it does	Price	PITLANE advantage
3D-printed card hooks (Etsy India)	Static hooks; cards hang flat, one behind another	₹277–302	Swing-out access, every card visible, rows step forward
Wall-mount 1:64 stands (direct sellers)	Printed stands for loose cars, packs of 12–48	₹550–1,690	Made for <i>carded</i> cars, double-locked clips
Modular 3D-printed wall case (Etsy India)	4 cars per module	₹1,658+	7 cars on one 40 cm strip for less
Display STL files (Etsy India)	Print-at-home files	₹1,188	Our STL pack at ₹499

Sources: [Etsy India: 3D-printed displays](#), [Etsy India: modular wall case](#), [Etsy India: display STL](#), [ModMuse wall stand](#), [eSUN PLA+ price in India](#), [Hot Wheels prices in India](#).

SWOT

Strengths

- A genuinely new mechanism: swing-out door for each card
- Every name visible; rows step out 7 mm
- No supports, low cost per unit, modular
- Designer is a collector, with real-print testing

Weaknesses

- Fits standard mainline cards only (~105 × 165 mm, blister ≤ 42 mm)
- Print time per kit is high (about 16 h for 4 cars, 26 h for 7)
- Ledges above must be opened to load a lower card
- The newest version (solid bump, L-lock) is not yet test-printed

Opportunities

- Sell the STL files too: zero marginal cost
- Colour editions (orange track, black and white)
- Premium / long-card ledge, Matchbox and Mini GT variants
- Hobby-shop wall displays (B2B)

Threats

- Cheap static hooks at ₹277–302
- Copycats on file marketplaces
- Trademark: never use “Hot Wheels” as your brand, only “fits Hot Wheels® cards”
- Courier cost of a 40 cm part (ship the 2-piece strip)

WHO BUYS IT

- **Adult carded collectors** (18–40, metro and Tier-1 cities) with 20–200 cards and a wall to fill. This is the main ad audience.
- **Parents and gift buyers:** birthdays, plus the Diwali and Christmas gift season (Oct–Dec).
- **Makers** with their own 3D printer, who buy the STL files.

COST TO MAKE (ESTIMATE, IN RUPEES)

Item	4-car kit	7-car kit	Basis
Filament (PLA+ at ₹1,500/kg)	₹297	₹474	From the STL files: about 198 g / 316 g printed (70% of fully solid)
Electricity	₹20	₹32	150 W printer, ₹8/kWh, 16 h / 26 h of printing at about 12 g/h
Failed-print allowance (15%)	₹48	₹76	Reprints, nozzle wear
Box, 4 screws + wall plugs, label	₹60	₹60	Ship the 2-piece strip to keep the box small
Total cost	≈ ₹425	≈ ₹640	Excludes your time and courier (about ₹80–120 within India)

RECOMMENDED PRICES (INR)

STL files ₹499 All parts and kits, print at home. Launch offer ₹399.	Starter (4 cars) ₹999 40 cm strip + 4 clips + 4 ledges. Room to add 3 more.	BEST VALUE Full strip (7 cars) ₹1,499 Fully loaded: ₹214 per car. Launch offer ₹1,299.	Add-on rack ₹199 One clip + one ledge. 3 for ₹549.
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Why these prices: static hooks sell for ₹277–302 and a 4-car modular case for ₹1,658. PITLANE sits above the hooks as a *functional premium* product but undercuts the case on price per car. Sell the 7-car strip at ₹1,499 with free shipping. Sold direct (Instagram or WhatsApp, paid by UPI), it leaves about **₹750 per kit** after cost and courier. On Amazon or Flipkart, with about 20% in fees, it leaves about **₹460**. Don't go below ₹1,199 for the full strip: it starts to look like a cheap hook. **Free shipping on orders of ₹999 and up;** ₹79 below that.

GO-TO-MARKET PLAN (INDIA)

- Week 0–2:** test-print the newest parts. Film a 15-second Reel of a ledge swinging open (the hook for every ad) and take photos on a real wall.
- Launch:** sell direct on Instagram and WhatsApp with UPI payment. Post in Hot Wheels collector groups on Facebook and WhatsApp. Give 3–5 local collectors and YouTubers a free strip for honest reviews.
- Marketplaces:** Amazon.in and Flipkart (search reach), Meesho (no commission), Etsy India (STL files and export orders).
- Meta ads:** ₹300–500/day, ages 18–40, metro and Tier-1 cities, interests: Hot Wheels, die-cast, Matchbox, car culture. Test the 5 posters for a week, keep the 2 with the cheapest clicks, and retarget profile visitors with the "7 cars ₹1,499" ad. Push hardest from October to December (Diwali and Christmas gifting).
- Listing title:** "PITLANE Swing-Out Wall Display for Carded 1:64 Die-cast Cars, fits Hot Wheels® & Matchbox® mainline cards, 7-car 40 cm strip".

NAME

PITLANE: every car gets its own garage door, like a pit box in the pit lane at a race, and the strip is the pit wall. The orange stripe nods to the orange track every collector remembers. Backups: **CardGarage**, **Swing Grid**. Before you register a name, check trademark and domain availability (India: the IP India trademark search, classes 20 and 28; plus Instagram handle and domain).